Physiology of Pregnancy

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Block: Endocrine/Urology/Reproduction

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Textbook

Chapter 3 in Gabbe, Obstetrics: Normal and Problem Pregnancies 7th Edition, China, 2017. Elsevier

Learning Objectives

After participating in this session, learners should be able to

- Describe specific physiologic changes to the cardiovascular, hematologic, pulmonary, renal, gastrointestinal, and endocrine system that occur in pregnancy
- Correlate the physiologic changes to all systems in pregnancy with the needs of the fetus and mother
- Apply the system-specific physiologic changes in pregnancy to explain normal and abnormal symptomatology in pregnancy

Study Questions

1. In a normal pregnancy the primary change in the cardiovascular system is a marked increase in cardiac output. What's the major cause of increased cardiac output in the first half of pregnancy?

a. Increased heart rate
b. Increased systemic vascular resistance
c. Cardiac stimulation by hCG
d. Increased stroke volume

2. What hormone causes the decrease in systemic vascular resistance during pregnancy?

a. Progesterone
b. Estrogen
c. Vasopressin
d. Human chorionic gonadotropin

3. Lowered hemoglobin during pregnancy can be a normal physiologic finding mainly due to:

a. Blood loss due to the placenta
b. Increased plasma volume
c. Increased cardiac output resulting in greater red cell destruction
d. Increased gastrointestinal excretion of iron

4. The Patient's blood volume in a normal pregnancy:

a. Remains stable
b. Increases by 5-10%
c. Increases by 40-50%
d. Decreases by 20%

5. Pulmonary function changes dramatically in pregnancy. Which statement is true?

a. Tidal volume increases and respiratory rate remains unchanged
b. Respiratory rate increases and tidal volume drops
c. Diaphragm lowers and O2 consumption diminishes
d. Respiratory acidosis and alkalosis occur simultaneously

A. Cardiovascular System

1. WHAT IS CARDIAC OUTPUT? SV X HR= CO
2. Cardiac output increases about 50% during pregnancy, and typically peaks at about 25-30 weeks, in the early third trimester.
3. By 5 weeks gestation, the CO has already risen 10%
4. Most of the increased cardiac output is going to the uterus, placenta, and breasts.
5. Both cardiac output and stroke volume increase in pregnancy primarily because of the drop in peripheral vascular resistance.
6. The decrease in peripheral vascular resistance is likely related to progesterone mediated smooth muscle dilation…but other factors are also likely involved, including the nitric oxide pathway and the blunting of responsiveness to vasoconstrictors, such as angiotensin II and norepinephrine.
7. The combination of increased stroke volume and increased heart rate can sometimes give patients the sensation that they are having palpitations. Although pregnancy does cause a slight increase in premature atrial and ventricular contractions, it is typically very reasonable to reassure patients that there is no pathologic process occurring.
8. Normal physiologic changes of pregnancy that can mimic disease include
    a. Dyspnea
    b. Pre-Syncope
    c. Decreased exercise tolerance

B. Hematologic Changes

1. 1 The patient's blood volume begins to increase it about 6 weeks gestation. It then rises progressively until 30-34 weeks and plateaus until delivery. The average expansion of blood volume is 40-50%. The range, however, can be anywhere between 20 and 100%.
2. Multiple gestations have a larger increase in blood volume than singleton pregnancies.
3. The increase in blood volume results from combined expansion of both plasma volume and red blood cell mass.
4. The increase in plasma volume occurs very early in the pregnancy; however, the red blood cell mass increase occurs more slowly.

5. This discrepancy creates a phenomenon that we call physiologic anemia of pregnancy.
6. Platelets stay in the normal range or slightly at the end of pregnancy. Gestational thrombocytopenia is common, and is characterized by platelet counts in the 140,000-70,000 range. The mechanism for this likely involves an immunologic response as well as increased platelet consumption
7. The white blood cell counts typically rises progressively in pregnancy. White counts in the 20-30 range are not uncommon during the labor and delivery process.

C. Pulmonary Changes

1. During pregnancy the mucosa of the upper respiratory tract becomes edematous and hypersecretory. This can lead to chronic nasal stuffiness and postnasal drip.
2. The configuration of the rib cage changes early in pregnancy. The chest expands, and the level of the diaphragm rises.
3. Lung volumes increase in pregnancy. By and large, the respiratory rate remains unchanged.
4. Tidal volume increases by 30-50% by 8 weeks gestation. The increase in tidal volume, coupled with the unchanged or slightly increased respiratory rate, creates a state of chronic mild hyperventilation. This results in increased alveolar oxygen and decreased arterial carbon dioxide.
5. The pregnant patient's oxygen consumption will be between 20 and 40% above nonpregnant levels. This occurs as a result of the rising oxygen requirement of the fetus and placenta as well as the increased oxygen requirement for maternal organs. Overall, this results in a lowering of the maternal oxygen reserve. This can often give a sensation of shortness of breath or even apnea.

D. Renal/Urinary System Changes

1. The kidneys increase in size during pregnancy. This is primarily due to an increase in the renal vasculature.
2. The ureters and renal pelvis also enlarges, and this is likely due to smooth muscle relaxation caused by progesterone.

3. Renal plasma flow increases markedly in early pregnancy. By 16 weeks gestation the effective renal plasma flow is 75% greater than nonpregnant levels. The glomerular filtration rate is approximately 50% higher than the nonpregnant state.
1. Creatinine clearance in pregnancy is greatly increased, because of increased
4. renal plasma flow and the increased glomerular filtration rate. Importantly, the increase in renal function resulting in hyperfiltration occurs without an increase in glomerular pressure, and this preserves the kidneys from damage.
5. As a result of glomerular hyperfiltration serum levels of creatinine, BUN, and uric acid all drop.
2. Glucose excretion increases in almost all pregnancies, but potassium is spared.

E. Body Water Metabolism

1. There is a total body water increase of 6.5-8.5 L by the end of gestation. This is one of the most impressive adaptations of pregnancy.
2. The additional water is accounted for by expansion of the blood volume, the plasma volume, and the red cell mass seen with pregnancy. There are also increases in extracellular fluid, intracellular fluid, amniotic fluid, and fluid distributed in the breasts and adipose tissue.
3. The increase in body water contributes to l pregnancy weight gain, hemodilution, the physiologic anemia of pregnancy, and the elevation of cardiac output.
4. Interestingly, sodium metabolism is delicately balanced and maintained secondary to enhanced tubular sodium reabsorption. Alterations in sodium handling represent the largest renal adjustment that occurs in gestation.
5. Hormonal control of sodium balance is under the opposing actions of the renin angiotensin-aldosterone system and the natriuretic peptides secreted by the heart.
6. The physiologic function of the renin-angiotensin-aldosterone system is dramatically increased in pregnancy, and likely originates from a drop in the mean arterial pressure that is influenced of gestational hormones and increased nitric oxide production. The renin system functions to maintain intravascular volume.

7. Simultaneously atrial and brain natriuretic peptide production by the heart acts as diuretics, natriuretics, vaso-relaxants, and overall antagonists to the reninangiotensin-aldosterone system.

F. Gastrointestinal Tract

1. Appetite is generally increased a bit in pregnancy.
2. Pregnancy reduces the risk for peptic ulcer disease, but gastroesophageal reflux is present in 30-50% of individuals. The increase in GERD is multifactorial, and likely comes from relaxation of the gastroesophageal sphincter and dysmotility. Later in the pregnancy, pressure on the stomach from the enlarging uterus likely contributes.
3. Small and large intestine motility is decreased in pregnancy. This results in constipation for many patients.
4. Gallbladder function is diminished. This results in an enlarged gallbladder for many patients, as well as increased chance of the development of gallstones.
5. The size and histology of the liver are for all intents and purposes unchanged in pregnancy. Liver function tests are also generally unaffected except in disease processes.
6. The etiology of morning sickness is poorly understood. It typically occurs between 8 and 12 weeks of gestation. A small percentage of patients will have continued morning sickness until the 20 to 22nd week. An even smaller percentage will suffer throughout the pregnancy until the time of delivery. Interestingly, patients with morning sickness have pregnancies with very favorable outcomes.

G. Gestational Weight Gain & Nutritition

3. Pregnancy remains a generalized period of weight gain for virtually everyone.
4. Normal weight gain is generally between 22 and 37 pounds.
5. In the modern era, recommendations regarding weight gain have changed considerably.
6. The Institute of Medicine (IOM) published specific guidelines in 2009 with concrete recommendations that are to some degree data driven and reflect

the general changes in body mass index that have occurred over the last 20-25 years.

a. BMI less than 18.5 kg/m^2 28-40 pounds
b. Normal BMI 18.5-24.9 kg/m^2 25-35 pounds
c. Overweight BMI 25-29.9 kg/m^2 15-25 pounds
d. Obese BMI 30-39.9 kg/m^2 11-20 pounds
e. Severe obesity BMI >40 kg/m^2 0-11 pounds

7. Calorie Increases
8. Micronutrient Requirements

H. Endocrine System

1. Thyroid disease is common in people of reproductive
2. Thyroid binding globulin increases early in pregnancy, as does total T4.
3. TSH drops initially, then returns to the normal level. The transient decrease in TSH coincides with a first trimester increase in free T4. Both effects appear to be caused by the thyrotropic effects of hCG.
4. Adrenocortical function increases in pregnancy. There are increased serum levels of aldosterone, corticotropin releasing hormone, ACTH, corticosteroid binding globulin, cortisol, and free cortisol. This causes an overall state of physiologic hypercortisolism.
5. In pregnancy, corticotropin releasing hormone is produced by the placenta and fetal membranes and is secreted into the pregnant patient's circulation, thus driving increased adrenal production of cortisol.

I. Pancreas and Fuel Metabolism

1. Pregnancy allows for the continuous transport of energy in the form of glucose to the baby.
2. This creates an overall deterioration in glucose tolerance for the pregnant patient, with a general trend toward a diabetogenic state.
3. Plasma lipids and lipoproteins increase in pregnancy. Total cholesterol and triglycerides are much higher than in the nonpregnant state. There is no reason to check for hypercholesterolemia or hyperlipidemia in pregnancy, since no treatment is indicated.

J. Musculoskeletal System

1. Pregnancy was initially believed to be a state of overall calcium loss, but this has not proven to be the case.
2. Total calcium levels decline throughout a person's pregnancy, however the physiologically important fraction, which is called serum-ionized calcium, is unchanged and constant. Serum calcium levels are maintained, and fetal calcium requirements are met through increased intestinal calcium absorption.
3. Pregnancy is a period of high bone turnover and remodeling. Although pregnancy and breast-feeding may cause transient bone loss, this is reversible. Studies do not support an association between parity and osteoporosis later in life.
4. Pregnancy is associated with two musculoskeletal changes that can result in significant discomfort for patients:
    - Increasing anterior convexity of the lumbar spine, which is known as lordosis, is a compensatory mechanism that keeps the center of gravity over the legs and prevents the enlarging uterus from shifting the center of gravity anteriorly.
    - Widening of the pubic symphysis occurs later in the pregnancy and this commonly results in pain near the symphysis that is often referred down the inner thighs and may cause al sensation of snapping or movement in the bones while walking.

K. Skin/CNS/Eyes

1. Skin: Skin changes revolve primarily around increased cutaneous blood flow and increased pigmentation because of increased production of melanocyte stimulating hormone by the placenta.
2. CNS: The pituitary gland enlarges during pregnancy.
3. Ocular changes sometimes occur during pregnancy. The two most significant of these are increased thickness of the cornea and decreased intraocular pressure. As a result of this, many patients have changes in their visual acuity during pregnancy. Many patients will need to have changes in their eyeglass and/or contact lens prescriptions following pregnancy.

L. Microbiome

1. Specific changes in the microbiome during pregnancy may serve to maintain pregnancy, prepare the body for delivery, and/or establish the neonatal microbiome at the time of delivery.
2. The vaginal microbiome changes to favor the lactobacillus species. These lactobacilli may play an important role in the prevention of infection and may help prevent preterm birth.
3. Contrary to popularly held medical belief, there is evolving evidence that suggests that the placenta and upper reproductive tract are not sterile and may contain helpful microbes. This research is in its infancy.

Answers to Study Questions

1. d
2. a
3. b
4. c
5. a